Module 2 — Read 1

Exact and approximate one-proportion inference

Module 2, Read 1: Exact and approximate one-proportion inference

Project: Project 2 — Estimating the Truth

Optional supplements :

- Textbook by **Vu & Harrington**, see Ch 3.3 — Normal distribution, Ch 4.1 — Variability in estimates, and Ch 4.2 — Confidence intervals;
- Videos: Khan: confidence intervals and margin of error · Khan: confidence interval simulation.
- ChatGPT or Notebook LM: give this page to a genAI and ask it to step you through concepts and test you with the questions to prepare for Quiz A. Ask it to cover each of the learning outcomes.

Start with **Read 0** then continue to Read 1. This reading is your self-paced textbook for Module 2, Part 1. Work one section at a time; use Check yourself boxes before Quiz A.

Learning outcomes by Section

Section 1: Type I and Type II errors

Type I and Type II errors in hypothesis tests

- Define Type I and Type II errors using the decision vs. reality table.
- Describe each error type in context for a one-proportion test.
- Relate the significance level α to the maximum acceptable Type I error rate.

Section 2: Exact binomial — `binom.test()`, tests vs confidence intervals

Exact binomial inference for one proportion

- Contrast a hypothesis test (*is* p equal to p_0?) with a confidence interval (*what* values of p are plausible?).
- Use `binom.test()` for an exact binomial p-value and confidence interval in R.
- Interpret a 95% CI for a proportion in context and compare with test conclusions.

Section 3: Approximate methods for one proportion (CLT, `prop.test`)

Normal approximation and Wald / `prop.test` for one proportion

- Use $\mu = np$ and $\sigma = \sqrt{np(1-p)}$ and explain why approximations were used historically.
- Check $np \geq 10$ and $n(1-p) \geq 10$ before normal-based methods; otherwise use exact or Plus Four.
- State the Empirical Rule; compute Wald and Plus Four 95% CIs; use `prop.test(x, n)` for approximate CIs when conditions hold.

Section 4: Standardized test statistic for one proportion

Standardized z for a one-proportion test

- Compute $z = (\hat{p} - p_0)/\sqrt{p_0(1-p_0)/n}$ (equivalently from count x).
- Connect standardized z to rejection using ± 2 on the standard normal scale.
- Contrast test SE (p_0) with CI SE ($\hat{p}$); relate **`prop.test()`** p-value and CI output.

Section 1: Type I and Type II errors

Type I and Type II errors in hypothesis tests

The trial-by-jury analogy

Hypothesis testing works like a trial by jury:

- Null hypothesis (H_0) = the default assumption of innocence. We assume this statement to be true until there is strong evidence otherwise.
- P-value = the weight of the evidence against the null. A small p-value means the data would be unlikely if H_0 were true, that is, strong evidence against "innocence." A large p-value means the data are plausible under H_0 so we do not have enough evidence to reject the default assumption.
- Reject H_0 = like a guilty verdict: we conclude the evidence is strong enough to reject the default assumption of innocence.
- Fail to reject H_0 = like a not-guilty verdict: we do not have enough evidence to reject the default; we are not saying H_0 is "true," only that we cannot reject it.

Performing a hypothesis test

In general, testing a claim can be done in 4 steps:

1. **State the hypotheses:** H_0 (null) and H_a (alternative) and draw the distribution of the statistic assuming the null hypothesis H_0 is true.
2. **Collect data** and compute the observed value of the statistic (e.g. sample proportion or count).
3. **Compute the p-value:** the probability of observing data as or more extreme than what we saw, ifH_0 were true.
4. **Decide:** If the p-value is small (e.g. below a chosen significance level α), reject H_0; otherwise fail to reject H_0.

Every decision is made in the face of uncertainty: H_0 might really be true or false, and we might reject it or not. Sometimes we make decisions that are incorrect, just like a jury might find an innocent person guilty.

Decision vs. reality

Our decision (reject or fail to reject H_0) can be right or wrong depending on reality (whether H_0 is actually true or false):

	Reality: H_0 true	**Reality:** H_0 false
Decision: Reject H_0	Type I error (false alarm)	Correct — we reject when we should (power)
Decision: Fail to reject H_0	Correct — we keep the default when it's true	Type II error (missed opportunity)

- **Type I error:** Rejecting H_0 when H_0 is actually true.
- **Type II error:** Failing to reject H_0 when H_0 is actually false.
- **Power:** Probability of correctly rejecting H_0 when H_0 is false. So $\mathbf{Power = 1 - P(Type\ II\ error)}$.

Defining Type I error, Type II error, and power

- **Type I error:** Rejecting the null hypothesis when it is actually true. We control this with the significance level α (e.g. $\alpha = 0.05$): we reject H_0 only when the p-value $< \alpha$.
- **Type II error:** Failing to reject the null hypothesis when it is actually false. Its probability depends on the true state of nature (e.g. how far the true p is from the null value), the sample size, and α.
- **Power:** The probability of correctly rejecting H_0 when H_0 is false. Power = $1 -$ P(Type II error). Higher power means we are more likely to detect a real effect when it exists.

Example 1: The "lucky" coin

Setup: We want to know if a coin lands on heads more often than not (a "lucky" coin for heads).

State the hypotheses

- H_0: The coin is fair; $p = 0.5$ (probability of heads).
- H_a: The coin is lucky for heads; $p > 0.5$.

Define errors in context

- **Type I error:** Concluding that the coin is "lucky" when it actually is not (false alarm).
- **Type II error:** Failing to conclude that the coin is "lucky" when it actually is (missed opportunity).

Define power

- Power = probability of correctly concluding the coin is lucky when it really is ($p > 0.5$); i.e. Power = $1 -$ P(Type II error).

Balance the errors

- Choose α (e.g. 0.05) to control how often we falsely declare the coin lucky. To reduce the chance of missing a truly lucky coin, increase the number of flips (sample size) to increase power.

Example 2: The "lucky" die

Setup: We want to know if a die lands on a particular side (e.g. 6) more often than it should.

State the hypotheses

- H_0: The die is fair; $p = 1/6$ (probability of that side).
- H_a: The die is "lucky" for that side; $p > 1/6$.

Define errors in context

- **Type I error:** Concluding that the die is "lucky" when it really isn't.
- **Type II error:** Failing to conclude that the die is "lucky" when it really is.

Define power

- Power = probability of correctly concluding the die is lucky when it really is; Power = $1 -$ P(Type II error).

Balance the errors

- Choose α to limit false claims of a lucky die. Increase the number of rolls to improve power if we care about not missing a biased die.

Example 3: Medical treatment effectiveness

Setup: A new medical treatment is studied. We want to know if it is effective in more than 10% of cases.

State the hypotheses

- H_0: The treatment is effective in exactly 10% of cases; $p = 0.10$
- H_a: The treatment is effective in more than 10% of cases; $p > 0.10$.

Define errors in context

- **Type I error:** Concluding that the treatment is effective in more than 10% of cases when it actually isn't (false discovery). The consequence of this error is that we adopt an ineffective treatment.
- **Type II error:** Failing to conclude that the treatment is effective when it actually helps more than 10% of patients (missed opportunity). The consequence of this error is that we abandon a beneficial treatment.

Define power

- Power = probability of correctly concluding the treatment is effective when it really is (e.g. when true $p > 0.10$); Power = 1 − P(Type II error).

Balance the errors

- If falsely approving an ineffective treatment is very serious, use a smaller α (e.g. 0.01). If missing a beneficial treatment is very serious, design the study with larger sample size (and possibly a somewhat larger α) to achieve higher power.

Increasing power

- Increase the sample size; optionally use a slightly larger α if Type II error is the main concern; or focus the alternative (e.g. a specific $p > 0.10$) to design for that effect size.

Summary of terms

Term	Meaning
Null hypothesis (H_0)	The null model; a statement of no effect or no difference; the default assumption (e.g. "innocence").
Alternative hypothesis (H_a)	The statement we consider if we reject H_0 (e.g. there is an effect or a difference).
P-value	The probability of observing data as or more extreme than what we saw, if the null model were true. Small p-value = strong evidence against H_0; large p-value = we cannot confidently reject H_0.
Type I error	Rejecting the null hypothesis when it is actually true (false alarm).
Type II error	Failing to reject the null hypothesis when it is actually false (missed opportunity).
Power	Probability of correctly rejecting H_0 when H_0 is false; Power = 1 − P(Type II error).

Use the trial-by-jury analogy: H_0 = default assumption of innocence, p-value = weight of evidence, and the Decision vs. Reality table to see how Type I, Type II, and power fit together.

Check yourself — Type I and Type II errors

1. Coin test $H_0 : p = 0.5$ vs $H_a : p > 0.5$: define **Type I** and **Type II** errors in context.
2. What is **power** in terms of Type II error?

Answers — Type I and Type II errors

1. Type I: conclude lucky when fair. Type II: miss a lucky coin.
2. Power $= 1 - P(\text{Type II})$ — probability of correctly rejecting a false H_0.

Section 2: Exact binomial inference

Exact binomial inference: tests vs confidence intervals

> **i** Bridge from Module 1
>
> In Module 1 you computed p-values with **`pbinom()`** after using **`dbinom()`** to find the bar heights in the visualization of how likely each value of x is if the null hypothesis were true. The purpose of Module 1 was to gain a deep understanding of the mathematical model behind the p-value. Now that you have this understanding, we'll use the R function `binom.test()` to get the same p-value. An additional reason for using this function is that it also gives us an exact confidence interval. It doesn't provide the visualization of comparing our observed value of x with the distribution under H_0 so we'll have to rely on the one number summary of this picture, the p-value.

Two different questions about one proportion

In project 1, you tested a claim about the value of the probability of success p using a p-value, $P(X \text{ as extreme as } x|H_0)$ and the binomial distribution. Another way to use the binomial distribution is to compute a **confidence interval**. This is a statistical tool that answers a slightly different question:

Statistical tool	Result	Example wording of the research question
Hypothesis test	p-value from exact or approximate test	*Is* the population proportion p **different from** (or above/below) a claimed p_0?
Confidence interval	Interval of plausible values	*What is* the population proportion p? Which values of p are **consistent with** the data?

- **Hypothesis test:** fixes a null value for p, p_0 and asks whether the data $\hat{p} = x/n$ are surprising if p_0 were true.
- **Confidence interval:** does not fix p at one null value but instead reports a range of values for p that are compatible with the observed data $\hat{p} = x/n$.

Note that both tools use the same binomial model (so we'll have to check the same 4 conditions for a BRP before using either) but they answer different research questions.

`binom.test()` — one-line replacement for hand-built `pbinom`

Module 1 workflow: H_0, H_a → plot `dbinom(...)` with observed x → shade tail(s) → `pbinom(...)` for p-value.

Module 2 workflow: H_0, H_a → `binom.test(...)` for both p-value and confidence interval

```
# Two-sided test: is p different from 0.5? 7 successes in 16 trials
binom.test(x = 7, n = 16, p = 0.5)
```

The output includes:

- **p-value** = 0.8036 which means there is an 80% chance of seeing the number of successes in 16 trials be as extreme as 7 if the underlying probability of success is 0.5. Because this p-value is large, I conclude the data is consistent with the null hypothesis that he probability of success is 0.5.

- **95% confidence interval** for the true proportion p is (0.1975341, 0.7012231). This can be interpreted as "I am 95% confidence that the underlying probability of success p is between 0.198 and 0.701.
- **estimate** $\hat{p} = x/n = 0.4375$

Example 1: Test Fair coin? Data 7 heads in 16 flips

- **Question:** $H_0 : p = 0.5$ vs $H_a : p \neq 0.5$.
- `binom.test(7, 16, 0.5)` → extract p-value (not probability of success) and compare p-value to $\alpha = 0.05$.
- **Interpretation:** If p-value is larger than 0.05, fail to reject H_0 and conclude data are consistent with a fair coin. If small, conclude data is not consistent with a fair coin.

Example 2: What is the probability of heads? Data 7 heads in 16 flips

- **Question:** What is the long-run proportion of heads for this coin?
- `binom.test(7, 16)` → extract interval endpoints.
- **Interpretation (CI):** *I am 95% confident that the true probability of heads on this coin is between 0.198 and 0.701.*

The **test** asked whether the data 7/16 is consistent with a particular value of p, in this case $p = 0.5$. The **CI** lists all values of p that are consistent with the data 7/16.

Example 3: Estimate the overall morality rate if there were 3 deaths in 16 patients

Estimate a hospital's mortality rate p with 95% confidence if there were 3 deaths in 16 patients.

- $\hat{p} = 3/16 = 0.1875$.
- **Exact:** `binom.test(3, 16)`

I am 95% confidence that the overall mortality rate for all patients at the hospital is between 0.04 and 0.46 based on data from 16 patients.

Summary

Tool	Module 1	Module 2
Null distribution bars	`dbinom` + `geom_col`	—
Tail probability / p-value	`pbinom`	`binom.test(..., p = p0)`
CI for p	—	`binom.test(..., conf.level = 0.95)`

Check yourself — Exact binomial — `binom.test()`, tests vs confidence intervals

1. 16 flips, 7 heads, coin toss: identify **parameter** p vs **statistic** $\hat{p}$.
2. 7 heads in 16 flips: what question does a **test** of $H_0 : p = 0.5$ answer? What question does a **95% CI** answer?

3. which R function gives the p-value and confidence interval from a binomial model?
4. Write one sentence interpreting a 95% CI interval (L, U) for a coin's p in context.

Answers — Exact binomial — `binom.test()`, tests vs confidence intervals

1. **Parameter** p = true long-run proportion of heads (**unobserved**). **Statistic** $\hat{p} = 7/16 = 0.4375$ (**observed**).
2. Test: is $p = 0.5$ plausible? CI: what range of p values is plausible?
3. `binom.test()` (exact binomial).
4. I am 95% confident the true probability of heads is between [L] and [U].

Note on notation: parameter p vs statistic $\hat{p}$

- **Parameter** p = true population proportion (**Greek / theoretical / unobserved**).
- **Statistic** $\hat{p} = x/n$ = sample proportion (**from data / observed**).
- `binom.test(x, n, p = p0)` tests a claim about p; `binom.test(x, n)` (no `p`) estimates p with a CI.

Bridge: From Exactly Right to Approximately Correct

Section 2 used exact `binom.test()` for tests and CIs. Section 3 estimates p with **approximate** Wald / `prop.test()` CIs and uses an approximately correct distribution to compute a p-value.

	Hypothesis test	**Confidence interval**
Question	Is p equal to a claimed p_0?	What values of p are plausible?
Statistical tool, sample conclusion	p-value, reject / fail to reject	Interval endpoints, interpret with "I am 95% confident…"
Exactly Right	`binom.test(x, n, p = p0)` → p-value	`bino.test(x, n)` → CI
Approximately Correct	`prop.test(x, n, p = p0)` → approximate p-value	`prop.test(x, n)` → approximate CI

Section 3: Approximate inference for one proportion

Approximate inference for one proportion

In the old days: why approximate methods?

Exact binomial math is always correct for BRP data, but before computers people often **approximated** the binomial with a **normal curve** when sample size n was large enough.

We still need the binomial center and spread to build that normal curve:

Binomial mean and standard deviation

> Bridge from Module 1
>
> You already know the binomial pmf from Module 1 Read 1. Here we summarize the **center** ($\mu = np$) and **spread** ($\sigma = \sqrt{np(1-p)}$) of the count X these will be the same μ and σ used when we approximate a binomial by a normal curve.

For a binomial random variable X with n trials and success probability p ($X \sim \text{Binomial}(n, p)$):

Mean (expected value):

$$\mu = np$$

Standard deviation:

$$\sigma = \sqrt{n\,p\,(1-p)}$$

Interpretation

- Mean μ: The center of the distribution of X. If we observed X many, many times (each time from a new binomial process with the same n and p), the average of those values would be $\mu = np$.
- Standard deviation σ: A typical deviation of X from the center. It is the square root of the average squared distance of (many) observations of X from the mean μ. So σ answers: "Roughly how far from μ do we expect a single observation of X to fall?"

Examples: Identify n and p, then compute μ and σ

For each scenario, (1) state what X counts and verify the setting is a binomial random process (binary outcome, fixed n, fixed p, independent trials); (2) identify n and p; (3) compute the mean $\mu = np$ and the standard deviation $\sigma = \sqrt{np(1-p)}$.

Example 1: ESP test

Suppose we run an ESP test with 25 trials. On each trial the person guesses one of 5 symbols; the computer picks one at random. So by chance the probability of a correct guess on any one trial is $p = 1/5 = 0.2$. Let X = number of correct guesses out of 25.

- X counts: number of correct guesses in 25 trials.
- **Binomial?** Binary (correct or not), fixed $n = 25$, fixed $p = 0.2$, independent trials. Yes.
- n and p: $n = 25$, $p = 0.2$.
- **Mean:** $\mu = np = 25 \times 0.2 = 5$.
- **SD:** $\sigma = \sqrt{np(1-p)} = \sqrt{25 \times 0.2 \times 0.8} = \sqrt{4} = 2$.

So we expect a person to get about 5 correct by chance, with a typical deviation of about 2.

Example 2: Left-handed students

Suppose we collect data from 30 students from our school. About 10% of people are left-handed nationwide so we assume $p = 0.10$ as the probability that a randomly chosen student is left-handed. Let X = number of left-handed students in the sample.

- X counts: number of left-handed students in 30 students.
- **Binomial?** Binary (left-handed or not), fixed $n = 30$, fixed $p = 0.10$, independent students. Yes.
- n and p: $n = 30$, $p = 0.10$.
- **Mean:** $\mu = np = 30 \times 0.10 = 3$.
- **SD:** $\sigma = \sqrt{np(1-p)} = \sqrt{30 \times 0.10 \times 0.90} = \sqrt{2.7} \approx 1.64$.

We expect about 3 left-handed students, with a typical deviation of about 1.64.

Example 3: Defective items

Suppose that 12% of items are defective. We inspect 20 items at random. Let X = number of defective items among the 20.

- X counts: number of defective items in 20 inspected.
- **Binomial?** Binary (defective or not), fixed $n = 20$, fixed $p = 0.12$ (under the supplier's claim), independent items. Yes.
- n and p: $n = 20$, $p = 0.12$.
- **Mean:** $\mu = np = 20 \times 0.12 = 2.4$.
- **SD:** $\sigma = \sqrt{np(1-p)} = \sqrt{20 \times 0.12 \times 0.88} = \sqrt{2.112} \approx 1.45$.

In 20 items, we expect about 2.4 defectives, with a typical deviation of about 1.45.

Example 4: Free throws

A basketball player shoots 40 free throws. Her long-run probability of making any single free throw is 0.75. Let X = number of made free throws out of 40.

- X counts: number of made free throws in 40 attempts.
- **Binomial?** Binary (make or miss), fixed $n = 40$, fixed $p = 0.75$, independent shots. Yes.
- n and p: $n = 40$, $p = 0.75$.
- **Mean:** $\mu = np = 40 \times 0.75 = 30$.
- **SD:** $\sigma = \sqrt{np(1-p)} = \sqrt{40 \times 0.75 \times 0.25} = \sqrt{7.5} \approx 2.74$.

We expect about 30 makes in 40 throws, with a typical deviation of about 2.74.

In each case, μ is the center of the distribution (the average of many observations of X), and σ is a typical deviation from that center.

Graphs: exact binomial vs. normal approximation

Blue bars = exact binomial probabilities $P(X = x)$ from `dbinom()`. **Red curve** = normal density with the **same** μ and σ from `dnorm()`. We draw both on the same x-axis (count X).

Vertical lines (Empirical Rule on the normal curve):

Line	Meaning
Red solid at μ	Center of the distribution
Orange dashed at $\mu \pm \sigma$	About **68%** of a true normal lies between these
Purple dotted at $\mu \pm 2\sigma$	About **95%** lies between these — the "$\pm$2 SD" rule behind many approximate CIs and tests

When the red curve **hugs** the blue bars, the normal approximation is **good**. When it misses (especially at the tails), prefer **`binom.test()`** from Section 2.

Example A: Poor approximation ($n = 10$, $p = 0.2$)

Setting: X = number of heads in **10** flips of a fair coin; test reference $p = 0.2$.

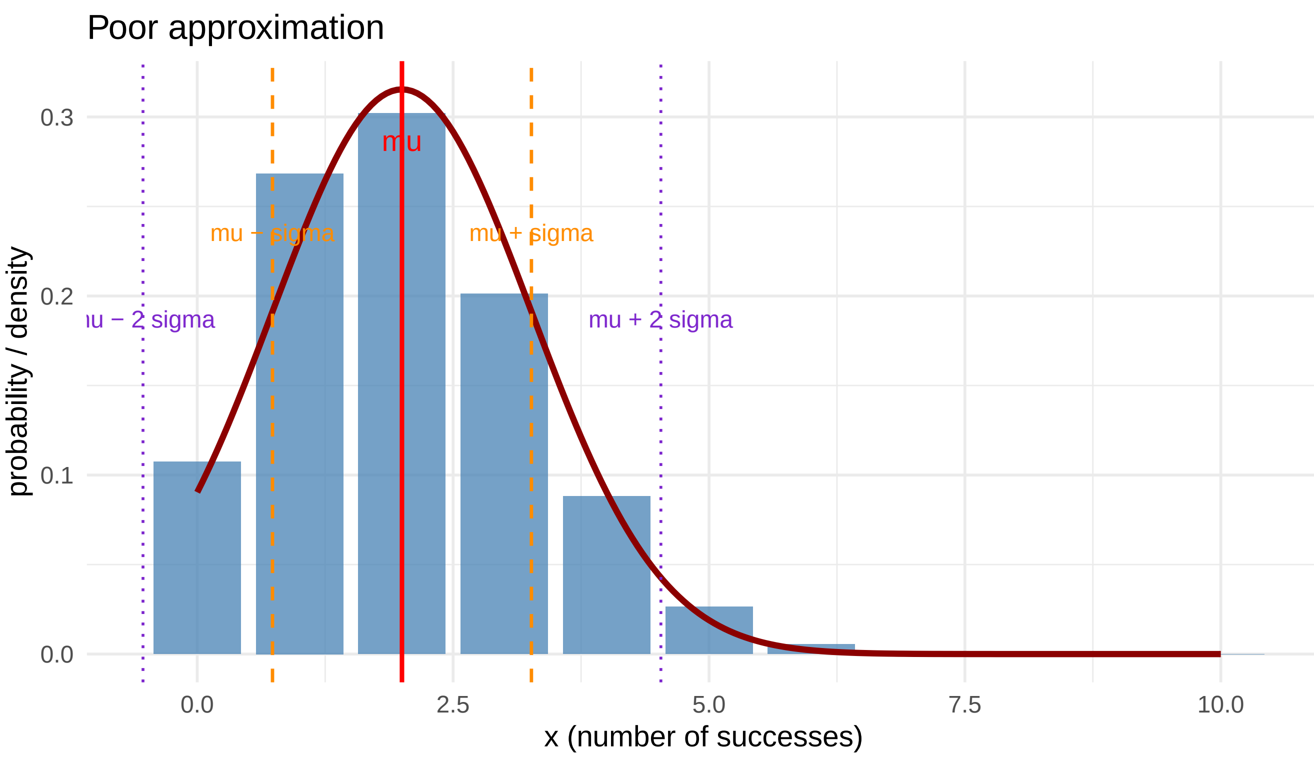

Figure 1: Poor normal approximation: n = 10, p = 0.2. Blue bars = exact binomial; red curve = normal with the same mu and sigma. Vertical lines mark mu and mu +/- 1 SD and +/- 2 SD.

The red curve does **not** match the bar heights well. Do **not** trust normal-based p-values or $\pm 2\sigma$ rules here, instead use **`binom.test()`**.

Example B: good approximation ($n = 50$, $p = 0.4$)

Setting: X = number of successes in **50** trials with $p = 0.4$.

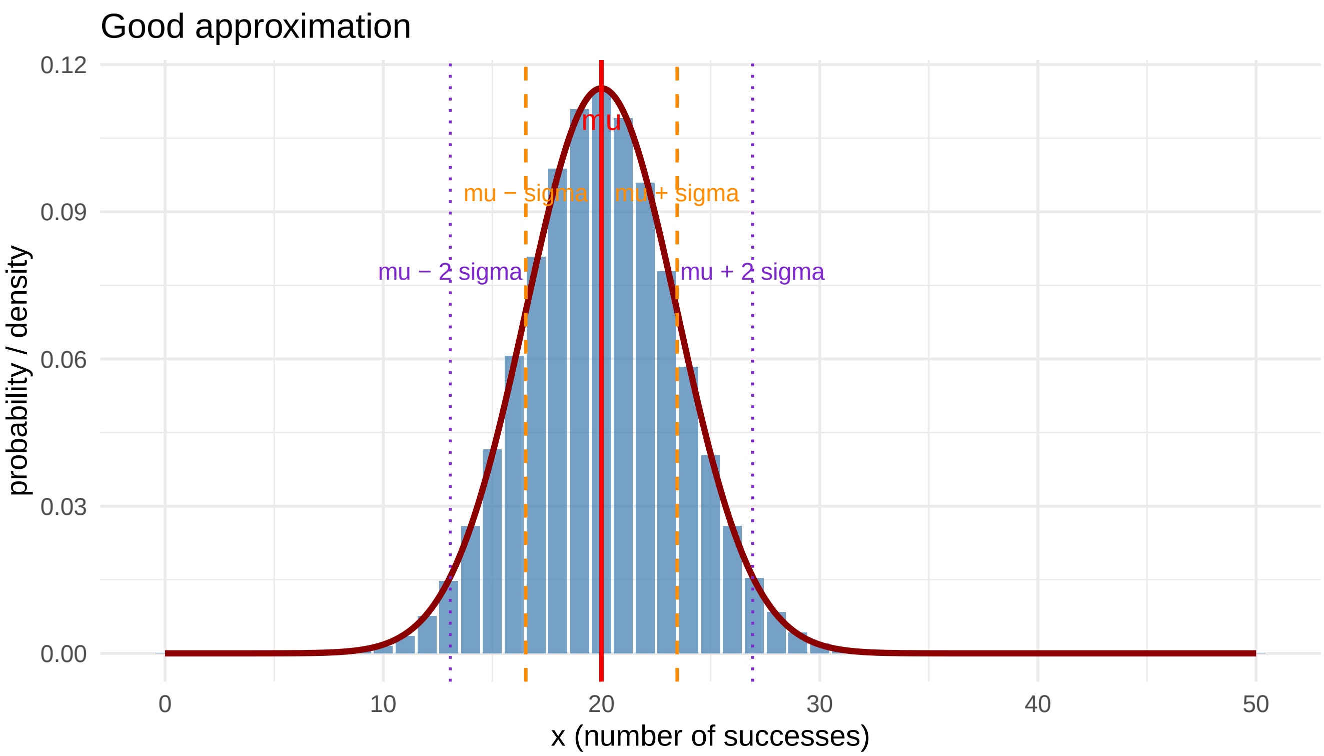

Figure 2: Good normal approximation: n = 50, p = 0.4. Bars follow the normal curve; mu +/- 1 SD and +/- 2 SD mark the Empirical Rule bands.

The bars sit under the bell curve and are close in the y direction.

Normal approximation to the binomial (The Central Limit Theorem)

When $X \sim \text{Binomial}(n, p)$ with n large enough, the distribution of X is approximately **normal** with the same mean and standard deviation:

- **Mean:** $\mu = np$
- **Standard deviation:** $\sigma = \sqrt{n\,p\,(1 - p)}$

So we can approximate $P(X \leq x)$ or $P(X \geq x)$ using a normal curve with that μ and σ.

Conditions for the approximation to be reasonable: Both the expected number of successes and the expected number of failures under the null are at least 10:

- $n\,p \geq 10$
- $n\,(1 - p) \geq 10$

This keeps the binomial from being too skewed so the normal curve fits reasonably well.

Empirical Rule (normal distribution)

For a **normal distribution** with mean μ and standard deviation σ:

- About **68%** of the distribution lies within **1** standard deviation of the mean: between $\mu - \sigma$ and $\mu + \sigma$.
- About **95%** lies within **2** standard deviations: between $\mu - 2\sigma$ and $\mu + 2\sigma$.
- About **99.7%** lies within **3** standard deviations: between $\mu - 3\sigma$ and $\mu + 3\sigma$.

Visualize with the One Proportion applet

Use the **One Proportion** applet to see the binomial distribution and its normal approximation

One Proportion applet (compare binomial bars to the normal overlay)

Try different values of n and p: when $np \geq 10$ and $n(1-p) \geq 10$, the normal curve overlays the binomial bars closely. When those conditions fail, the binomial is skewed and the normal fit is poor.

Conducting a two-sided test by hand

To test a null hypothesis about a proportion (e.g. $H_0 : p = 0.5$ for a fair coin):

1. Under H_0, identify n and p; compute $\mu = np$ and $\sigma = \sqrt{np(1-p)}$.
2. Sketch a **normal (bell shaped) curve** centered at μ with spread σ and mark μ, $\mu \pm \sigma$, $\mu \pm 2\sigma$ using the Empirical Rule.
3. Mark the **observed value** of x (the count) on the horizontal axis.
4. **Shade the rejection region**: if the level of significance $\alpha = 0.05$ then this region is above $\mu + 2\sigma$ or below $\mu - 2\sigma$.
5. **Conclude** by rejecting the null hypothesis H_0 if x is in the rejection region, e.g. reject H_0 if $x > \mu + 2\sigma$ or $x < \mu - 2\sigma$. If x is between $\mu - 2\sigma$ and $\mu + 2\sigma$, then conclude the data is consistent with H_0.
   Note: the **p-value** is the area of the shaded region beyond x under the curve. It is not possible to compute this by hand but we can use the above method to determine if the p-value is larger or smaller than 0.05, which is enough to make a conclusion.

Example 1: Testing a coin (reject H_0)

Research question: Is the coin fair? We test $H_0 : p = 0.5$ vs $H_a : p \neq 0.5$, where p = probability of heads on a single flip.

Data: We flip the coin **100** times and observe **35** heads.

- n and p under H_0: $n = 100$, $p = 0.5$.
- **Check conditions:** $np = 50 \geq 10$ and $n(1-p) = 50 \geq 10$. OK to use the normal approximation.
- **Mean and SD:** $\mu = 100 \times 0.5 = 50$, $\sigma = \sqrt{100 \times 0.5 \times 0.5} = 5$. 68% of the values of X are expected to be between 45 and 55; 95% are expected to be between 40 and 60.
- **Observed:** $x = 35$ heads.
- **Rejection region:** Shade both below 40 and above 60 to find the rejection region for a test with $\alpha = 0.05$.
- **Conclusion:** the data 35 is below 40 so **Reject** H_0. The data are inconsistent with a fair coin; we have evidence that $p \neq 0.5$ (here we saw fewer heads than expected 95% of the time if the coin were fair).

Example 2: Testing a die (fail to reject H_0)

Research question: Is the die fair for the face “6”? We test $H_0 : p = 1/6$ vs $H_a : p \neq 1/6$, where p = probability of a 6 on a single roll.

Data: We roll the die **60** times and observe **12** sixes.

- n and p under H_0: $n = 60$, $p = 1/6$.
- **Check conditions:** $np = 60 \times (1/6) = 10 \geq 10$ and $n(1-p) = 60 \times (5/6) = 50 \geq 10$. OK to use the normal approximation.
- **Mean and SD:** $\mu = 60 \times (1/6) = 10$, $\sigma = \sqrt{60 \times (1/6) \times (5/6)} = \sqrt{50/6} \approx 2.89$.
- **Observed:** $x = 12$ sixes.
- **Rejection region:** For $H_a : p \neq 1/6$, we would shade both above 10+2(2.89) = 15.78 and below 10-2(2.89)=4.22. Th
- **Conclusion: Fail to reject** H_0 because the data 12 is between 4.22 and 15.78. The data are consistent with a fair die; we do not have evidence that $p \neq 1/6$.

We can also obtain an approximate 95% confidence interval for p using the Empirical Rule and the Central Limit Theorem. This can be done by estimating the value of p in the formula for σ by substituting in $\hat{p}$ and then doing a bit of algebra to convert counts to proportions.

Formula 1: One proportion (Wald formula)

For a **population proportion** p, we estimate it with the sample proportion $\hat{p}$. The standard deviation of $\hat{p}$ is:

$$\mathrm{SE}(\hat{p}) = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

So the **95% confidence interval for a single proportion** is:

$$\hat{p} \pm 2\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

Conditions for Formula 1 (Wald):

1. **Binomial random process:** (a) **B**inary outcome (success/failure), (b) **I**ndependent trials, (c) **N**umber of trials n fixed in advance, (d) **S**ame probability of success π on each trial.
2. **Normal approximation:** $n\hat{p} \geq 10$ and $n(1-\hat{p}) \geq 10$ (at least 10 successes and 10 failures in the sample, so the sampling distribution of $\hat{p}$ is approximately normal).

Example 1: Confidence interval for a proportion (St. Joe's) — large sample

Goal: Estimate the mortality rate at St. Joe's (Eureka, CA) for this heart procedure using the **large sample** (200 patients, 14 deaths), with **95% confidence**.

- **Data:** $n = 200$, $x = 14$ deaths.
- **Statistic:** $\hat{p} = 14/200 = 0.07$ (7%).
- **Standard error:**
$$\mathrm{SE}(\hat{p}) = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}} = \sqrt{\frac{0.07 \times 0.93}{200}} \approx \sqrt{0.0003255} \approx 0.0180.$$
- **95% CI:** $\hat{p} \pm 2 \times \mathrm{SE}(\hat{p}) = 0.07 \pm 2(0.0180) = 0.07 \pm 0.036$.
- **Interval:** Lower bound ≈ 0.034, upper bound ≈ 0.106, so about [**0.034, 0.106**] or [**3.4%, 10.6%**].

Interpretation: We are 95% confident that the true mortality rate for this heart procedure at St. Joe's is between about 3.4% and 10.6%.

Small sample for a proportion: Plus Four

When the sample size is small, the condition for the normal approximation may not be satisfied. Then:

- For a **95% confidence interval** for a proportion, use the **Plus Four** formula.
- If the confidence level is not 95% and there are at least 10 successes and 10 failures in the sample, you may use the Wald (z) formula. Otherwise, consult a statistician.

Plus Four formula: Add 2 "successes" and 2 "failures" to the sample. With x = number of successes and n = sample size, define

$$\tilde{p} = \frac{x+2}{n+4}$$

and use the **effective sample size** $n+4$. The standard error of $\tilde{p}$ is

$$\mathrm{SE}(\tilde{p}) = \sqrt{\frac{\tilde{p}(1-\tilde{p})}{n+4}}.$$

The **95% confidence interval for a proportion (Plus Four)** is:

$$\tilde{p} \pm 2\sqrt{\frac{\tilde{p}(1-\tilde{p})}{n+4}}.$$

Example (small sample): Estimate the mortality rate at St. Joe's using the **small sample** (**16** patients, **3** deaths) with 95% confidence. Use the Plus Four method.

- **Data:** $x = 3$ deaths (successes), $n = 16$ patients.
- **Plus Four estimate:** $\tilde{p} = \frac{x+2}{n+4} = \frac{3+2}{16+4} = \frac{5}{20} = 0.25$.
- **Standard error:** $\mathrm{SE}(\tilde{p}) = \sqrt{\frac{\tilde{p}(1-\tilde{p})}{n+4}} = \sqrt{\frac{0.25\times0.75}{20}} = \sqrt{0.009375} \approx 0.0968$.
- **95% CI:** $\tilde{p} \pm 2 \times \mathrm{SE}(\tilde{p}) = 0.25 \pm 2(0.0968) = 0.25 \pm 0.194$.
- **Interval:** Lower bound ≈ 0.056, upper bound ≈ 0.44, so about [**0.06, 0.44**] or [**6%, 44%**].

Interpretation: We are 95% confident that the true mortality rate for this heart procedure at St. Joe's is between about 6% and 44%. With only 16 patients, the interval is wide and the estimate is uncertain; the Plus Four method gives a plausible range that is more reliable than the Wald interval for such a small sample.

Check yourself — Approximate methods for one proportion (CLT, `prop.test`)

1. Coin: $n = 10$, $p = 0.5$. Compute μ and σ by hand. Do normal-approx conditions pass?
2. $n = 50$, $p = 0.4$. Compute μ, σ, and check conditions.
3. On the overlay plot, what do the purple dotted vertical lines mark?
4. St. Joe's: $n = 200$, $x = 14$ deaths. Compute $\hat{p}$ and approximate 95% Wald CI.
5. St. Joe's small sample: $n = 16$, $x = 3$. Compute Plus Four $\tilde{p}$ and approximate 95% CI.
6. `prop.test(x, n)` gives?

Answers — Approximate methods for one proportion (CLT, `prop.test`)

1. $\mu = 5$, $\sigma = \sqrt{2.5} \approx 1.58$. $np = 5$ and $n(1-p) = 5$ — **both fail**; poor approximation.
2. $\mu = 20$, $\sigma = \sqrt{12} \approx 3.46$. $np = 20$ and $n(1-p) = 30$ — **both pass**; good approximation.
3. $\mu \pm 2\sigma$ — about **95%** of a normal lies between them (Empirical Rule).
4. $\hat{p} = 0.07$. CI $\approx [0.034, 0.106]$.
5. $\tilde{p} = (3+2)/(16+4) = 0.25$; SE ≈ 0.097; CI $\approx [0.06, 0.44]$.
6. **Approximate CI** for p

Section 4: Standardized test statistic for one proportion

Standardized test statistic for one proportion

Standardizing statistics used to conduct a hypothesis test

In the optional reading below, you'll see the **z-score** as a way to compare relative standing in two different groups or within a group, assuming the reference group is normally distributed.

Let's apply the same idea of shifting and scaling to conducting a hypothesis test of a population proportion or probability p.

If $np \geq 10$ and $n(1-p) \geq 10$, then the normal distribution can be used to approximate the **binomial** distribution of $X =$ number of successes in n trials, where p is the probability of success in a single trial and the trials are independent.

We used this fact along with the empirical rule that 95% of the values of a normal distribution are within 2 SDs of the mean. We computed the mean of X as $\mu = np$ and the SD of X as $\sigma = \sqrt{np(1-p)}$, and then rejected the null hypothesis if our observed value x was below $\mu - 2\sigma$ or above $\mu + 2\sigma$.

Another way to conduct this test is to **standardize** the observed value x by subtracting the mean and dividing by its standard deviation (also called **standard error** in this context):

Definition: standardized test statistic

$$\frac{\text{observed statistic} - \text{mean of sampling distribution}}{\text{standard deviation of sampling distribution}}$$

In the case of the statistic being a count of successes X from a binomial random process with n and p, the standardized test statistic is

$$z = \frac{x - np}{\sqrt{np(1-p)}}.$$

This value can be compared with the standard normal distribution which has a mean of $\mu = 0$ and a standard deviation of $\sigma = 1$; a null hypothesis is rejected if z is below -2 or above 2 (because the standard normal distribution has mean 0 and SD 1 so $\mu \pm 2\sigma = \pm 2$).

Equivalently, a standardized test statistic can be computed using the observed proportion $\hat{p} = x/n$: it is the same as the one based on x, but all terms are divided by n:

$$z = \frac{\hat{p} - p}{\sqrt{p(1-p)/n}}$$

Note that p is the value of the population proportion or probability hypothesized under the null hypothesis, and $\hat{p}$ is the observed proportion from n trials.

Example: Test whether a coin is fair if 16 heads were observed in 20 flips using a significance level of 0.05. Use a standardized test statistic and assume it follows the standard normal distribution.

> **Approximate Test by hand**
>
> Under $H_0 : p = 0.5$, $np = 10$ and $n(1-p) = 10$. $z = \dfrac{16 - 20(0.5)}{\sqrt{20(0.5)(0.5)}} = \dfrac{6}{\sqrt{5}} \approx 2.68$. This is beyond ± 2, so we **reject** H_0 at $\alpha = 0.05$.

Same standardizing idea but tests use p_0, CIs use $\hat{p}$ for the SE

The z formulas below are for hypothesis tests: the standard error uses the null p_0 because you ask whether the data are surprising if $p = p_0$.

For a 95% CI (estimation), the Wald / **`prop.test(x, n)`** interval uses $\hat{p}$ in the SE instead:

$$\mathrm{SE}(\hat{p}) = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}, \qquad 95\%\ \mathrm{CI} \approx \hat{p} \pm 2 \cdot \mathrm{SE}(\hat{p}).$$

So: same "observed minus reference, divided by SE" pattern; different reference (p_0 for tests, $\hat{p}$ for CIs). When large-count conditions hold, the p-value from `prop.test(x, n, p = p0)` and the CI from `prop.test(x, n)` are two views of the same normal approximation (Section 3).

Check yourself — Standardized test statistic for one proportion

1. Write the standardized statistic for testing $H_0 : p = p_0$ using observed $\hat{p}$.
2. 16 heads in 20 flips, $H_0 : p = 0.5$: is z beyond ± 2?
3. For a Wald CI, which p goes in the SE , p_0 or $\hat{p}$?

Answers — Standardized test statistic for one proportion

1. $z = (\hat{p} - p_0)/\sqrt{p_0(1-p_0)/n}$ when large-count conditions hold.
2. $z \approx 2.68$ — yes, reject at $\alpha = 0.05$.
3. $\hat{p}$ — estimation uses the observed proportion, not the null.

Optional Reading (not required for Project 2)

- Normal distribution and z-scores — standardization for individuals and reference distributions
- Organizing the statistics toolbox — variable types and descriptive vs. inferential map
- Rubber-band CI analogy — intuition for confidence intervals